

EIE3123 DYNAMIC ELECTRONIC SYSTEMS: QUIZ 3

1. For the third-order servo system

$$G(s) = \frac{50,000}{s(s+10)(s+50)},$$

use Bode plot sketches to design a lead compensator so that phase margin (PM) $\geq 50^\circ$ and $\omega_{BW} \geq 20$ rad/sec. Verify and refine your design by using MATLAB.

2. For a system with open-loop transfer function

$$G(s) = \frac{10}{s[(s/1.4)+1][(s/3)+1]},$$

design a lag compensator with so that PM $\geq 40^\circ$. What is the approximate bandwidth of this system?